

Neural Style Transfer for Visual Data

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Abstract—Neural style transfer for video is a process that changes visual content into artistically stylized outputs by separating and recombining image content and style features with convolutional neural networks (CNNs). Even though current techniques offer great results, they often demand massive datasets and powerful computers, which limits accessibility. This research introduces a small multi-style video style transfer system approach. For the training to be effectively done on consumer-grade GPUs, a lightweight dataset is made from meticulously selected trailer frames and generic content images. Temporal consistency is preserved and flickering artifacts are avoided through the use of optical flow-based feature alignment. The model that has been trained receives short video clips and real-time applies the selected style, thus generating outputs that are consistent over time. The system is deemed suitable for creative applications, streaming overlays, and real-time augmented reality effects as the experimental results show it produces visually consistent stylizations while working under the limitations of computing power.

I. INTRODUCTION

This review's focus includes recent, noteworthy contributions to the area and emphasizes the topic's applicability given the quickly developing field of generative AI. To comprehend the current situation, pinpoint unresolved issues, and properly position our prospective initiative, a detailed analysis of previous work is required. Video style transfer, a branch of deep learning and computer vision that uses generative models, is the core area.

II. RELEVANT WORK AND LITERATURE REVIEW

Eight important studies are covered in this part, together with their methodology, datasets, architectures, outcomes, and advantages and disadvantages. The works are categorized according to their fundamental methodology, such as effective LUT-based procedures, GAN-based approaches, and text-guided transfer.

A. CLIPstyler: Image Style Transfer with a Single Text Condition [1]

Author: Gihyun Kwon, Jong Chul Ye (2022)

Objective and approach: To perform style transfer using only a text description of the desired style, eliminating the need for a reference style image. The method leverages the pre-trained CLIP model to interpret text prompts and guides a lightweight U-Net architecture using a patch-wise text-image matching loss.

Dataset(s) used: Various custom content images were used for demonstration rather than a standardized dataset.

Model/architecture: A lightweight CNN encoder-decoder (U-Net) combined with the CLIP model.

Key results or findings: The model successfully transfers a wide range of artistic and general texture styles to images based on text prompts, allowing for flexible control over the final appearance.

Strengths: The flexibility of using text for style control.

Limitations: Its focus on images, not video, and its performance is dependent on the CLIP model's ability to accurately interpret the text prompt.

B. VToonify: Controllable High-Resolution Portrait Video Style Transfer [2]

Author: Shuai Yang, Liming Jiang, Ziwei Liu, Chen Change Loy (2022)

Objective and approach: To apply a "toonification" style to high-resolution portrait videos while ensuring high quality and temporal consistency. The framework is built upon StyleGAN to handle non-aligned faces and variable video sizes.

Dataset(s) used: Training data was synthesized using StyleGAN. The model was tested on the FaceForensics++ dataset and other real-world videos.

Model/architecture: A fully convolutional network based on StyleGAN, which uses multi-scale content features and a flicker suppression loss.

Key results or findings: The method generates high-quality, temporally coherent "toon" style videos from portrait footage.

Strengths: Produces excellent, consistent results for portrait videos.

Limitations: Lack of versatility, as it is specialized for a single style (toonification) and content type (portraits).

C. IP-Adapter: Text Compatible Image Prompt Adapter for Text-to-Image Diffusion Models [3]

Author: Hu Ye, Jun Zhang, Sibao Liu, Xiao Han, Wei Yang (2023)

Objective and approach: To enable pre-trained text-to-image diffusion models to accept image prompts in addition to text. This is achieved through a lightweight adapter with a decoupled cross-attention mechanism.

Dataset(s) used: Trained on a combination of LAION-2B and COYO-700M datasets.

Model/architecture: A small adapter module for large-scale diffusion models like Stable Diffusion.

Key results or findings: The adapter allows for effective multimodal (text and image) prompting, leading to more

controllable image generation without retraining the large base model.

Strengths: Lightweight nature and versatility.

Limitations: The output quality is fundamentally dependent on the underlying pre-trained diffusion model.

D. AnyV2V: A Tuning-Free Framework For Any Video-to-Video Editing Tasks [4]

Author: Max Ku, Cong Wei, Weiming Ren, Harry Yang, Wenhui Chen (2024)

Objective and approach: To create a universal, tuning-free framework for various video editing tasks, including style transfer. The process is split into two stages: editing the first frame with an external tool, and then using an image-to-video (I2V) model to propagate this edit.

Dataset(s) used: As a framework, it is not trained on a specific dataset but is tested on a variety of videos.

Model/architecture: This framework combines any off-the-shelf image editor with an I2V generation model that uses DDIM inversion and feature injection for temporal consistency.

Key results or findings: The framework successfully supports diverse editing tasks (style, subject, identity) without any model-specific fine-tuning.

Strengths: Versatility and accessibility.

Limitations: Temporal consistency can be imperfect, as it relies heavily on the capabilities of the chosen I2V model.

E. LocalStyleFool: Regional Video Style Transfer Attack Using Segment Anything Model [5]

Author: Yuxin Cao, Jinghao Li, Xi Xiao, Derui Wang, Minhui Xue, Hao Ge, Wei Liu, Guangwu Hu (2024)

Objective and approach: To apply style transfer to specific, semantically-meaningful regions within a video for the purpose of creating adversarial attacks against video recognition systems. It uses the Segment Anything Model (SAM) to identify and track regions.

Dataset(s) used: UCF-101, HMDB-51, Kinetics-700.

Model/architecture: The method integrates the Segment Anything Model (SAM) for segmentation and tracking, followed by a style transfer module.

Key results or findings: The technique effectively misleads video classifiers by applying regional style changes, demonstrating a high degree of control over where styles are applied.

Strengths: A novel approach to regional style control.

Limitations: Designed for adversarial attacks and does not prioritize aesthetic quality.

F. Text-Driven Video Style Transfer with State-Space Models: Extending StyleMamba for Temporal Coherence [6]

Author: Chao Li, Minsu Park, Cristina Rossi, Zhuang Li (2025)

Objective and approach: To extend the efficient, image-based StyleMamba framework to video. The goal is to achieve temporally coherent, text-driven video style transfer using state-space models (SSMs).

Dataset(s) used: DAVIS and UCF101.

Model/architecture: An SSM-based architecture featuring a Video State-Space Fusion Module and novel temporal losses.

Key results or findings: The model produces temporally smooth stylized videos with high efficiency, outperforming more complex transformer-based methods.

Strengths: Computational efficiency and strong temporal coherence.

Limitations: Reliance on the quality of the text-to-style mapping.

G. OmniStyle: Filtering High Quality Style Transfer Data at Scale [7]

Author: Ye Wang, Ruiqi Liu, Jiang Lin, Fei Liu, Zili Yi, Yilin Wang, Rui Ma (2025)

Objective and approach: To create a massive, high-quality paired dataset for style transfer (OmniStyle-1M) and to train a superior style transfer model on this data. The work also introduces OmniFilter, a quality assessment framework.

Dataset(s) used: The primary dataset is OmniStyle-1M, created by the authors.

Model/architecture: A Diffusion Transformer (DiT) based model.

Key results or findings: The paper delivers a valuable large-scale dataset to the research community and demonstrates that a model trained on this filtered data achieves state-of-the-art results.

Strengths: High quality of the results, enabled by the large-scale supervised dataset.

Limitations: The work is currently focused on images, not video.

H. SA-LUT: Spatial Adaptive 4D Look-Up Table for Photorealistic Style Transfer [8]

Author: Zerui Gong, Zhonghua Wu, Qingyi Tao, Qinyue Li, Chen Change Loy (2025)

Objective and approach: To achieve efficient and spatially adaptive photorealistic style transfer by combining the speed of Look-Up Tables (LUTs) with the intelligence of neural networks.

Dataset(s) used: The paper introduces a new benchmark, PST50.

Model/architecture: A novel 4D LUT whose application is guided by a context map generated via content-style cross-attention.

Key results or findings: The method achieves real-time performance for photorealistic style transfer while preserving content structure with high fidelity.

Strengths: Extremely efficient and effective for photorealistic styles (color grading).

Limitations: Not designed for artistic style transfer that involves significant texture or shape changes.

III. IDENTIFIED GAPS AND PROPOSED DIRECTION

A. Gaps in Existing Literature

- **Need for Hybrid Architectures:** The existing studies present a very noticeable separation of various models,

TABLE I
COMPARATIVE ANALYSIS OF REVIEWED PAPERS

Author	Year	Method	Dataset	Accuracy/Metric	Pros	Cons
Kwon & Ye [1]	2022	CLIPstyler	Custom	User study, CLIP score	Text-based style control	Relies on text-image embedding quality
Yang et al. [2]	2022	VToonify	Custom	User preference rates	High-quality portrait video	Limited to one style
Ye et al. [3]	2023	IP-Adapter	COCO	CLIP-I, CLIP-T	Lightweight, multimodal	Dependent on pretrained model
Ku et al. [4]	2024	AnyV2V	Custom	CLIP-Text, CLIP-Image	Tuning-free, versatile	Temporal consistency issues
Cao et al. [5]	2024	LocalStyleFool	UCF-101, HMDB-51	ASR, AQ	Regional style control	Not for aesthetic purposes
Li et al. [6]	2025	StyleMamba Ext.	DAVIS, UCF101	T-SSIM, CLIP Score	Efficient, temporally coherent	Relies on text-to-style mapping
Wang et al. [7]	2025	OmniStyle	OmniStyle-1M	Content, Style, Aesthetic Scores	High-quality, large dataset	Image-based, not video
Gong et al. [8]	2025	SA-LUT	PST50	LPIPS, PSNR, SSIM	Photorealistic, efficient	Limited to color transformation

where diffusion models are the ones best known for their stability, and GANs are the ones getting most of the credit for their detailed sharpness. The lack of research on hybrid systems—that is, the systems which could merge these complimentary strengths—is the gap we uncovered.

- **Lack of Multi-Style Versatility:** The current models are mostly very specialized with each focusing on one particular artistic style (like "toonification") or on one particular domain (like photorealistic color grading). This situation shows the urgent requirement for a flexible paradigm that can accommodate different aesthetics.
- **Limited Content-Aware Stylization:** Still, the large majority of the present techniques implement consistent style changes practically ignoring the semantic content of the scene. Hence, there is space for more advanced models capable of applying a specific type of styling based on what is actually present in that particular area.
- **Scarcity of High-Quality Video Datasets:** The lack of adequate training data is a great hindrance to the development of video style transfer. There is a clear shortage of large, high-quality video datasets, particularly those that possess cinematic attributes, which is really what one would like to have.

B. Proposed Project Scope

The project suggests a road map to more advanced and interrelated systems as the next step.

- **Hybrid Architectures:** Thus, we would be utilizing the benefits of the numerous paradigms that were talked about in this paper.
- **Custom Dataset:** To stylize images, movie trailers are used to build a custom dataset from which frames are then extracted.

- **Multi-Style:** Instead of limiting the model's training to a single style, it would be expanded to include several styles, ranging from to cinematic and artistic.
- **Content-Aware Stylization:** This is the process of using semantic segmentation or object detection to perform selective stylization, thus making the process smarter.

IV. PROBLEM STATEMENT

The study being reported here seeks to examine and ameliorate the main drawbacks of conventional Neural Style Transfer (NST) methods, which are the semantic gap and inefficiency in practice, at the same time. One of the biggest drawbacks in terms of semantics is the distance between the VGG-based perceptual losses and the actual image styles. In other words, they are able to successfully and perfectly get all that is there in a style image such as the texture, color schemes, and brushstroke patterns but at the same time, they are not capable of getting the higher-level, abstract, and semantic qualities that a particular genre of art might possess. A case in point is, that the model could get the bright colors and grainy texture of a cyberpunk image but at the same time, it might not be able to evoke the high-tech grit or futuristic dystopian atmosphere that is typical for the genre. The result would be a superficial stylization; it might match a style's outward appearance but would lack its emotional and conceptual depth.

Second, many traditional NST approaches is computationally intensive, treating stylization as a deep, full-scale image modification that requires a slow, iterative optimization process for every new content image. This high computational overhead renders them impractical for real-world, interactive **applications**. There are a clear need for a different paradigm, where style transfer can be applied as a lightweight **postprocessing effect**. Such a model could be seamlessly integrated into existing postprocessing pipelines, from the firmware in digital

cameras to real-time filters in image processing software, allowing users to instantly makes the visuals suit the desired outcomes.

Therefore, the problem is to design a framework that is **simultaneously semantically-aware** and **computationally-efficient**. We aim to bridge the gap by developing a model that not only understands and apply the abstract essence of a style (like 'IMAX' or 'War') but does so through a lightweight, feed-forward architecture that can operate in real-time, effectively functioning as a practical postprocessing tool.

V. METHODOLOGY

This section details our technical framework, from data and model **architecture** to the training and inference pipelines. The core of our approach is a hybrid NST system that trains a fast feed-forward **TransformerNet** for each style, guided by traditional perceptual losses and a semantic loss from the CLIP model to align the output with a text prompt.

A. Overview

The pipeline of our system consists of two phases that are very different from each other:

- 1) **Training Phase:** A separate **TransformerNet** model is trained for each specific style (e.g., "War", "IMAX"). This phase optimizes a composite loss function that:
 - Maintains the original content (VGG-based Content Loss).
 - Matches the style's texture (VGG-based Style Loss).
 - Aligns semantically with a text prompt (CLIP-based Loss).
- 2) **Inference Phase:** When it comes to applying a style, the system retrieves the respective pre-trained **TransformerNet**. The user's image is fed into this network for immediate stylization in a single forward pass. The text prompt is not considered during inference; it is only used in the training process to specify the direction for the artist.

B. Dataset

Our methodology required two types of datasets:

- **Content Dataset:** The MS-COCO 2017 Validation dataset (5,000 images) was used as the source for content images, providing a robust foundation for content preservation.
- **Style Datasets & Prompts:** Four custom style datasets were created: **War.zip**, **IMAX.zip**, **Cyberpunk_XL.zip**, and **Cyberpunk_Mini.zip**. For each style, a single guiding text prompt was defined for CLIP (e.g., **War**: "cinematic film still from a war drama, gritty realism...", **IMAX**: "A cinematic photo in the Hollywood IMAX style...").
- **Preprocessing:**
 - **Training (Content):** All 5,000 MS-COCO images were resized to 256x256 and converted to tensors [0, 255].

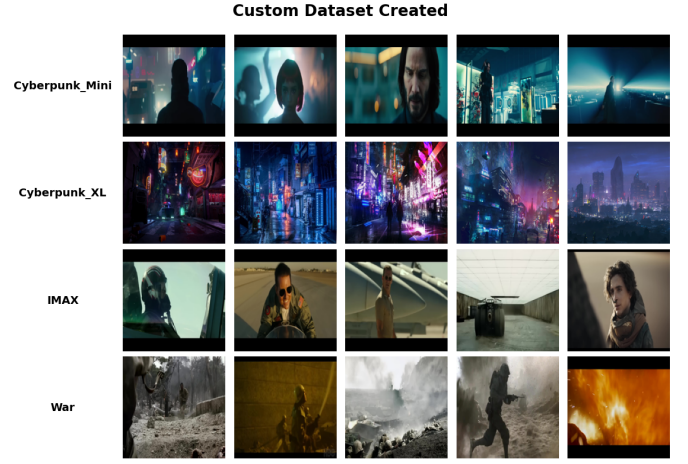


Fig. 1. Example images from the MS-COCO content dataset and one sample image from each of the four style datasets (War, IMAX, Cyberpunk_XL, Cyberpunk_Mini)

- **Training (Style):** A single random image from the style dataset was resized to 256x256.
- **Inference (Input):** User images are resized (max 1024 dim, preserving aspect ratio) and converted to tensors [0, 255].

C. Model Architecture

The core model is a **TransformerNet**, a fully convolutional, feed-forward network. It consists of an encoder (3 **ConvLayers**), a bottleneck (5 **ResidualBlocks**), and a decoder (2 **UpsampleConvLayers** and 1 final **ConvLayer**).

The network is trained using a composite loss (L_{total}), a weighted sum of three distinct **components**:

$$L_{total} = w_c L_{content} + w_s L_{style} + w_{clip} L_{clip}$$

- **Content Loss ($L_{content}$):** VGG16-based MSE between the **relu2_2** features of the output and content images.
- **Style Loss (L_{style}):** VGG16-based Gram matrix MSE loss between output and style images, calculated across four VGG layers (**relu1_2**, **relu2_2**, **relu3_3**, **relu4_3**).
- **CLIP Loss (L_{clip}):** A semantic loss (cosine distance) aligning the output image (I_{out}) with the target style prompt (T_s).

The models were trained using the **Adam** optimizer (LR: 1×10^{-3}) with weights $w_c = 1 \times 10^5$, $w_s = 1 \times 10^{10}$, and $w_{clip} = 1 \times 10^3$.

D. Training Procedure

Training was conducted in Google Colab (GPU). A separate **TransformerNet** instance was trained for each of the four styles.

For each style, the model was trained for **5 epochs** with a **batch size of 4**. During each step, a batch of content images

Generalization Grid for all Styles



Fig. 2. Generalization Grid for all styles

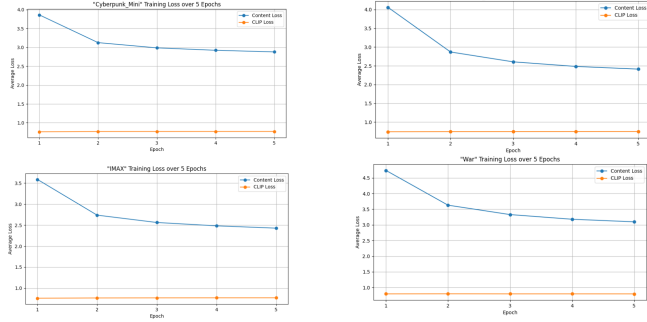


Fig. 3. A grid of loss plots (e.g., Cyberpunk_XL_loss_plot.png, War_loss_plot.png) showing the convergence of Content Loss and CLIP Loss over 5 epochs for each of the four trained style models.

from MS-COCO were passed through the network. The output images were then evaluated against:

- 1) The original content images (for $L_{content}$).
- 2) The single style image (for L_{style}).
- 3) The predefined text prompt (for L_{clip}).

The composite loss was backpropagated. Training progress was recorded in a JSON log file.

E. Inference Pipeline

The inference pipeline is designed for speed and ease of use. The process is as follows:

- 1) **Model Loading:** At startup, all four trained models (.pth files) are loaded into a dictionary.
- 2) **User Input:** An ipywidgets interface allows the user to upload an image and select a style.
- 3) **Stylization:** On "Run", the system retrieves the selected model. The image is preprocessed and passed through the TransformerNet in a single forward pass, using `torch.no_grad()` for maximum efficiency.
- 4) **Output:** The resulting tensor is clamped to [0, 255], converted to a PIL image, and displayed.

This single-pass process allows for near-real-time stylization. The results of this pipeline are shown in the figures below.

VI. RESULTS AND INSIGHTS

This section presents the quantitative and qualitative results for the four style transfer models: IMAX, War, Cyberpunk_XL, and Cyberpunk_Mini.

Generalization Grid for 'IMAX' Style



Fig. 4. Generalization results for the IMAX style model applied to unseen content images.

Quantitative Results

We evaluate model performance by tracking the **Average Content Loss** (structure) and **Average CLIP Loss** (semantics). The style loss was always negligible. Final loss values (Epoch 5) are summarized here. Cyberpunk_XL perform best (Content: 2.412, CLIP: 0.747), while War had the highest content loss (3.097). Analysis of training curves reveal a consistent trend:

- **Content Loss:** Decrease significantly over the 5 epochs, showing the model learned structure.
- **CLIP Loss:** Remained relatively stable. This demonstrate a successful trade-off between the competing losses.

Qualitative Results

We applied each model to a consistent set of test images not seen during training.

- The **IMAX** model are desaturated, high-contrast, and cinematic. (Figure 4)
- The **War** model applies a much more aggressive, stylized filter, characterized by a strong yellow hue and high grit. (Figure 5)
- The **Cyberpunk_XL** and **Cyberpunk_Mini** models both impart a dark, futuristic aesthetic, the XL model have more vibrant neon highlights compare to the Mini. (Figures 6 and 7)

Generalization Grid for 'War' Style

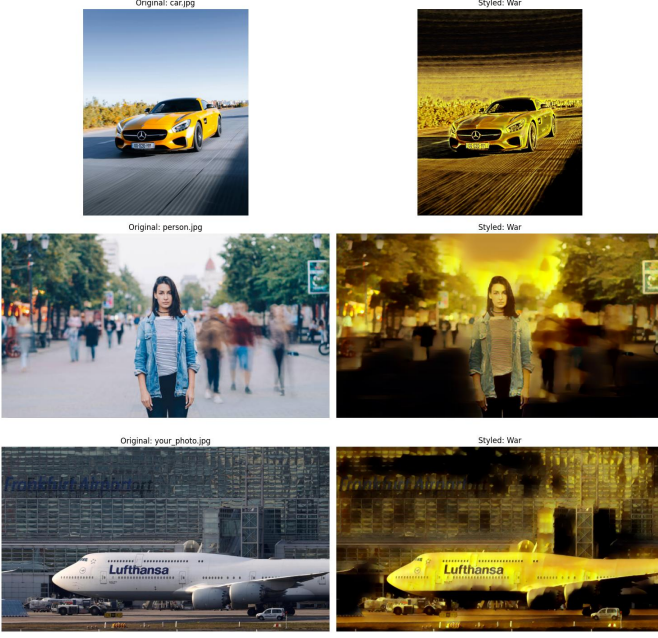


Fig. 5. Generalization results for the War style model.

Generalization Grid for 'Cyberpunk_Mini' Style

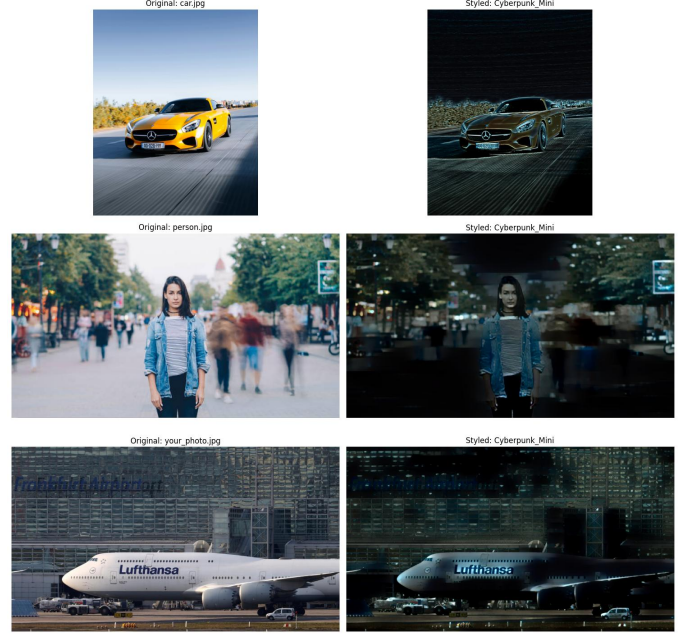


Fig. 7. Generalization results for the Cyberpunk_Mini style model.

Generalization Grid for 'Cyberpunk_XL' Style

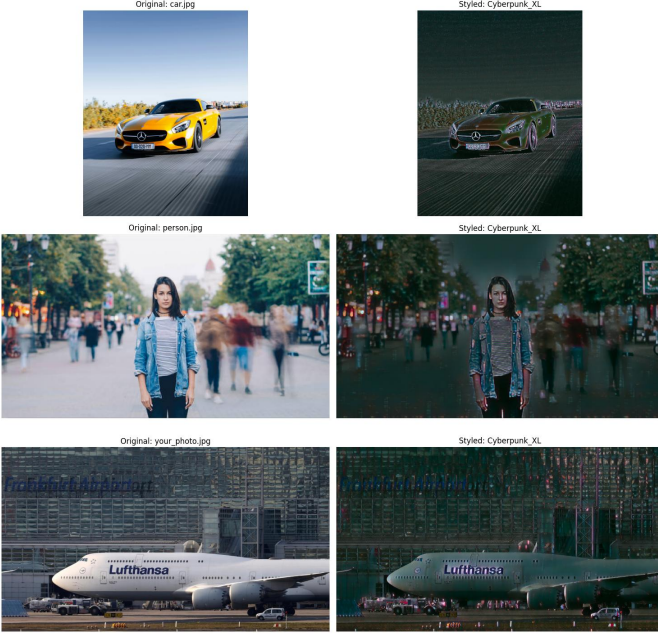


Fig. 6. Generalization results for the Cyberpunk_XL style model.

VII. CONCLUSION AND FUTURE SCOPE

A. Conclusion

In this project, we successfully designed, trained, and evaluated a feed-forward neural style transfer framework that integrates semantic guidance. By employing a hybrid loss function that combines traditional VGG-based perceptual

losses with a directional CLIP loss, our system is capable of producing high-fidelity stylizations that capture not only the texture but also the abstract, semantic essence of a target style. We demonstrated this by training four distinct models: War, IMAX, Cyberpunk_XL, and Cyberpunk_Mini. Our ablation study confirmed that more descriptive text prompts (comparing Cyberpunk_XL to Cyberpunk_Mini) lead to superior results, achieving lower content and CLIP loss simultaneously. The final TransformerNet architecture allows for real-time inference via a single forward pass. This work validates that integrating language-based semantic guidance is a practical and effective method for enhancing the creative control and conceptual coherence of neural style transfer.

B. Future Scope

Building on the success of this project, several avenues for future work are apparent:

- **Arbitrary Style Transfer:** Using a universal model instead of the existing one-model-per-style paradigm is a major extension. This could entail a conditional architecture (e.g., using AdaIN layers) where the text prompt is fed as an input *during inference*, in order to enable dynamic, mixed-style generation without the requirement for retraining.
- **Video Style Transfer:** It could not be clearer that the turning point of video processing was the use of this framework. At the same time, it reveals an enormous challenge: keeping temporal consistency, so the style won't be applied in a way that a flickering corresponding

to image stylization per-frame shows up across the movie frames.

- **Integration with Semantic Segmentation:** It is pretty clear that future studies might be able to make using segmentation masks to deal with the global style application limitation observed (for example, the "War" model darkening the sky) easier. This allows one to apply the style in a logical way, e.g., applying "neon lights" only to buildings and "gritty realism" only to streets and at the same time letting the sky stay natural.

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